

# Ryan Cho

Full Stack Software Engineer | Fixed Income & Analytics

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## PROFESSIONAL SUMMARY

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Full stack software engineer with seven years delivering business-critical applications in investment banking and asset management. Builds end-to-end analytics platforms, fixed-income valuation services, and modern web interfaces using Python, C#.NET, and TypeScript/React. Partners with portfolio managers, risk, and operations to replace legacy reporting with auditable microservices, SQL pipelines, and cloud tooling. Applies quantitative methods and AI-assisted development to ship reliable production systems.

## WORK EXPERIENCE

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### Software Engineer

May/2022 - May/2026

#### Aspen Capital

Remote | 10th Floor, New York

- Built **full-stack credit analytics platform** serving portfolio managers with React front end and **Python microservices** handling position ingestion, normalization, and reconciliation across private credit and structured product books
- Developed **fixed-income cash flow and valuation models** using pandas/NumPy for scenario analysis, duration/convexity sensitivities, and OAS-style spread analytics consumed by risk and investor relations teams
- Migrated **legacy SSRS and Excel-based reporting** to cloud-deployed REST APIs and responsive TypeScript/React dashboards, reducing ad-hoc turnaround for DDQ and portfolio composition reporting
- Integrated **Geneva portfolio accounting feeds and Bloomberg market data** through tested adapter services with validation rules, lineage tracking, and exception handling for firmwide position data quality
- Designed **asynchronous data pipelines and tuned SQL window-function queries** supporting daily NAV, exposure, and amendment monitoring across CLO and bank loan portfolios
- Led **requirements gathering with PMs, operations, and IR stakeholders**; authored functional specs, UAT plans, and deployment runbooks for analytics modernization initiatives under CIO-aligned technology roadmap
- Established **CI/CD pipelines on Azure** with unit/integration test coverage, Git-based PR reviews, and production monitoring across multiple concurrent delivery streams
- Built **Tableau and internal web reporting layers** for investor-facing portfolio summaries, credit deal lifecycle tracking, and underwriting-to-closing workflow visibility
- Adopted **AI coding assistants in daily workflow** to accelerate boilerplate reduction and test generation while enforcing code review, security, and verification standards before production release
- Owned **production support for business-critical analytics services**, triaging incidents, tuning query performance, and coordinating cross-team fixes with data management and business technology groups

### Technology Analyst, Incoming Technology Analyst, Technology

May/2019 - Apr/2022

#### Summer Analyst,

#### BNP Paribas

Remote | Seventh Avenue, New York

- Delivered **full-stack fixed income and structured products tooling** for trading and client solutions groups using C#.NET backend services and JavaScript front ends supporting pricing, risk, and operational workflows
- Built **REST APIs and asynchronous batch services** integrating market data vendors and internal pricing engines for credit instruments, bank loans, and derivative cash flow schedules
- Developed **SQL-heavy reporting modules with optimized window functions** supporting daily P&L, exposure roll-ups, and regulatory reporting across capital markets desks
- Contributed to **modernization of legacy .NET applications** toward modular microservice patterns with improved test coverage, source control practices, and DevOps-managed release cycles
- Implemented **quantitative analytics prototypes** for yield curve scenarios, spread decomposition, and cash flow projection using Python/pandas consumed by quantitative analysts and structurers

- Partnered with **non-technical stakeholders including portfolio managers and operations** to translate business requirements into technical specifications and lead integration testing across cross-functional teams
- Designed **responsive web interfaces for capital marketing analytics**, ensuring cross-platform usability for desktop and mobile users in fast-paced trading environments
- Supported **production deployment and incident response** for business-critical systems, maintaining high standards for data accuracy, operational monitoring, and post-release defect remediation

## Software Engineering Intern

**Jun/2018 - May/2019**

### Chexology

*Remote | 17th Floor, New York*

- Built **JavaScript/React features for customer-facing SaaS application** supporting restaurant compliance workflows, collaborating with product and design on responsive UI implementation
- Developed **Python REST API endpoints and unit tests** for backend services handling document validation and asynchronous notification processing
- Participated in **Git-based CI/CD pipeline improvements**, code reviews, and Agile sprint delivery across full-stack feature ownership from requirements through deployment
- Integrated **third-party payment and CRM adapters** with reconciliation logic, ensuring data consistency across operational dashboards used by internal teams
- Optimized **PostgreSQL queries and schema indexing** for high-volume transactional reads, improving API response times for production user traffic

## Web Developer

**Sep/2017 - Nov/2017**

### Bayesquare Foundation

*Remote | New York*

- Developed **responsive HTML5/CSS and JavaScript web pages** for nonprofit digital presence, ensuring cross-browser compatibility and mobile-friendly layout standards
- Implemented **contact and donation form integrations** with backend validation and third-party service hooks for operational data capture
- Collaborated with stakeholders to **gather content requirements and iterate on UX improvements** within tight nonprofit delivery timelines
- Maintained **version-controlled codebase** with structured commits and basic deployment workflow for site updates

## EDUCATION

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### New York University

*Bachelor of Arts (B.A.), Computer Science*

**2016-2020**

## SKILLS

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**Frontend:** React, TypeScript, JavaScript, Angular, HTML5, CSS3, Responsive Web Design, Cross-Platform Mobile Optimization

**Backend:** Python, C#.NET, REST APIs, Asynchronous Services, Microservices, Node.js, SSRS Migration

**Databases:** SQL Server, PostgreSQL, Expert SQL (Window Functions, Query Tuning), MongoDB, Redis, Set-Based Data Modeling

**Cloud & DevOps:** Azure, AWS, CI/CD, Git, Docker, Azure DevOps, TFS, Infrastructure-as-Code

**Monitoring & Tools:** Tableau, NumPy, pandas, Bloomberg API, Geneva, Unit/Integration Testing, AI Coding Assistants, Application Performance Monitoring

**Workflow & PM:** Agile/Scrum, Confluence, JIRA, Azure DevOps, UAT Planning, Requirements Gathering, Technical Documentation, Stakeholder Communication